PatientTriage.ai: Business Proposal & Strategy

“Triage is a snapshot. Risk isn't.” — Transforming Emergency Safety & Throughput

Accenture Innovation Challenge 2026 • Round 2 Business Case & Phased Roadmap

NET ANNUAL ROI	AVOIDED ICU COLLAPSE	LWBS RECOVERY	TARGET MARKET
$3.82M / Hospital	**-64% Decompensation**	**+$1.12M Recovered**	5,500+ US/EU Emergency Depts

Executive Summary

Emergency department (ED) crowding is a global healthcare crisis with over 140M annual visits in the US alone. Traditional emergency triage operates on an outdated premise: **a single static snapshot taken at intake**. Once triaged, patients wait unmonitored for 3 to 6 hours; physiological decline may remain undetected until a subsequent reassessment or clinical deterioration becomes apparent. When unmonitored patients silently deteriorate, the results are catastrophic: preventable in-hospital cardiac arrests, unanticipated ICU transfers, heightened malpractice claims, and severe nurse burnout.

PatientTriage.ai delivers a continuous physiological safety layer that monitors vital velocity (ΔSpO■, ΔHR), dynamic evidence decay ($\tau_{staleness}$), data uncertainty (*Unknown is NOT Safe*), and physician coverage (**The Attention Gap**). The intelligence layer performs continuous trend analysis and prioritization, while deterministic safety rules provide hard guardrails. It generates **$3.82M in net annual value** per 500-bed hospital while eliminating silent waiting room mortality.

1. Problem Framing & Competitive Moat (Vs. Native EHR Scores)

- **Silent Decompensation:** ESI Level 3/4 patients worsen unmonitored; physiological decline may remain undetected until subsequent checks.
- **The 'Missing Data Is Safe' Myth:** Legacy systems treat missing vitals as benign. Under *Unknown is NOT Safe*, missing data heightens vigilance.
- **Competitive Moat vs. Native EHRs (Epic EDI / Cerner MEWS):** Native EHR scores are designed for admitted inpatients in hospital beds and only score clinical severity. PatientTriage.ai is purpose-built for the waiting room and uniquely accounts for **physician coverage** and **evidence staleness decay**.

2. Solution Design & 3-Tier Layered Architecture

PatientTriage.ai enforces a strict 3-tier architecture: **Tier 1: Deterministic Safety Layer** (hard red-flags and downgrade blocking); **Tier 2: AI Decision-Support Layer** (vital velocity, dynamic confidence decay, and Attention Gap re-ranking with multimodal BLE sensor and kiosk ingestion); and **Tier 3: Clinician Governance Layer** (clinician override authority with mandatory justification and append-only audit logging). Runs air-gapped on edge hardware with sub-15ms latency.

3. Target Users & Stakeholder Value Propositions

Stakeholder	Primary Clinical / Financial Pain Point	PatientTriage.ai Value Proposition
Triage Nurses (RNs)	Overwhelmed tracking 40+ waiting patients; fear of silent deterioration.	Surfaces Top 3 actionable tasks with single Next-Best-Action buttons.
Emergency MDs	Blind to which waiting patient has worsened since initial intake.	Attention Gap Queue dispatches doctors to highest unmet clinical need.
Chief Medical Officers	Delayed diagnosis lawsuits, sentinel events in waiting lounges.	100% Downgrade Guardrails & append-only audit trail for malpractice defense.
Hospital CFOs	Uncompensated ICU boarding, LWBS revenue leakage, nurse turnover.	Delivers measurable $3.82M annual net ROI per 500-bed hospital facility.

4. Business Case, Financial ROI & Impact Model (500-Bed Hospital)

Model based on 65,000 annual ED visits, 500 acute care beds, and $1,200 average ED revenue per visit:

Value Driver	Pre-Implementation Baseline	Post-Implementation Impact	Annual Financial Value
1. LWBS Revenue Recovery	3,120 patients/yr leave (4.8%)	30% reduction via proactive re-engagement	**+$1,123,200 / yr**
2. Avoided ICU Transfers	145 waiting room ICU crashes/yr	64% reduction (93 avoided ICU stays @ $15k)	**+$1,395,000 / yr**
3. Malpractice Risk Mitigation	$1.2M annual liability reserve	40% reduction via documented audit trail	**+$480,000 / yr**
4. Nurse Retention & Overtime	26.8% nurse turnover (14 replacements)	4 replacements avoided + 15% overtime reduction	**+$378,000 / yr**
5. ED Throughput & Boarding	248 mins average wait/boarding	30-minute reduction via optimized dispatch	**+$445,000 / yr**
TOTAL GROSS ANNUAL VALUE	—	—	**$3,821,200 / yr**
Software Subscription & Support	—	Enterprise Tier License	-$240,000 / yr
NET ANNUAL ROI TO HOSPITAL	—	**14.9x Net Return on Investment**	**+$3,581,200 / yr**

5. Default Attention Gap Coefficients & Formula Weights

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Action Priority = (w_r × Risk + Urgency) + (w_d × Deterioration) + (w_s × Staleness) + Wait Hazard + (w_u × Uncertainty) - (w_c × Clinical Coverage)
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Parameter Bounds: $w_r = 1.0$ • w_d = +25 to +40 pts (ΔSpO■ ≤ -5% / ΔHR ≥ +20 bpm) • w_s = +20 to +35 pts (expiry) • w_u = +15 to +25 pts (missing vitals) • w_c = -35 pts (when is_attended = True).

6. Phased Implementation Roadmap & Precise Trial Endpoints

Phase	Timeline	Milestones & Trial Endpoints
Phase 1: Lab Validation	Q3 2026	20 synthetic benchmark scenarios validated across 33 automated tests; sub-15ms inference latency.
Phase 2: Shadow Trial	Q4 2026	Silent shadow deployment alongside Epic/Cerner via HL7 FHIR; clinician concordance evaluation.
Phase 3: Live Hospital Pilot	Q1–Q2 2027	**Primary Safety Endpoint:** >45% reduction in Mean Time to Escalation (MTTE); **Operational Endpoint:** <2 non-actionable alerts/nurse/shift; **Economic Endpoint:** ≥25% LWBS reduction over 90 days.
Phase 4: Enterprise Scale	Q3 2027+	Regional hospital network rollout with multi-facility dashboarding and centralized telemedicine escalation.

7. Key Risks, Regulatory Compliance & Mitigations

- **AI Hallucination Risk:** Mitigated by 3-tier architecture where deterministic safety red-flags override all statistical models.
- **Clinician Alarm Fatigue:** Mitigated by queue compression surfacing only the top 3 actionable tasks rather than flooding nurses.
- **Regulatory Classification (SaMD):** Positioned as Non-Device CDS (21 U.S.C. § 360aaa-1); clinician retains 100% decision authority.
- **Data Privacy (HIPAA/GDPR):** Runs air-gapped on local hospital network with zero external cloud LLM dependencies and append-only audit logging.

Conclusion: PatientTriage.ai delivers the technological breakthrough hospitals urgently require: an intelligent, explainable, and ethically grounded continuous safety layer that protects patients, empowers clinicians, and unlocks millions in enterprise hospital value.